Tutorial 10: Support Vector Machines

Margin geometry, hard and soft margins, and the dual

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Task 1: Comparing linear separators

Four labeled two-feature observations are:

i	$\boldsymbol{x}_i = (x_{i1}, x_{i2})^\mathsf{T}$	y_i
1	$(3, 1)^\mathsf{T}$	$+1$
2	$(2, 2)^\mathsf{T}$	$+1$
3	$(0, 1)^\mathsf{T}$	-1
4	$(1, -1)^\mathsf{T}$	-1

Compare:

$$f_A(\boldsymbol{x}) = x_1 - \frac{3}{2}, \qquad f_B(\boldsymbol{x}) = 2x_1 + x_2 - \frac{7}{2}.$$

For each, $\boldsymbol{w}$ is the coefficient vector and b is the intercept.

(a) Compute all scores and verify the predicted classes.

(b) Compute $\gamma_{\min} = \min_i y_i f(\boldsymbol{x}_i)/\|\boldsymbol{w}\|_2$ for both separators. Which has the larger margin?

(c) Rescale f_B canonically, then give its two margin lines and full margin width.

(d) Identify the margin-contact points. Verify that scaling f_B by 4 preserves the geometric margin of observation 2.

Solution 1(a): Scores and predictions

(a) A positive score predicts $+1$; a negative score predicts -1.
For $f_A = x_1 - 3/2$ and $f_B = 2x_1 + x_2 - 7/2$:

i	y_i	$f_A(\boldsymbol{x}_i)$	$f_B(\boldsymbol{x}_i)$	Prediction (both)
1	$+1$	$3 - \frac{3}{2} = \frac{3}{2}$	$6 + 1 - \frac{7}{2} = \frac{7}{2}$	$+1$
2	$+1$	$2 - \frac{3}{2} = \frac{1}{2}$	$4 + 2 - \frac{7}{2} = \frac{5}{2}$	$+1$
3	-1	$0 - \frac{3}{2} = -\frac{3}{2}$	$1 - \frac{7}{2} = -\frac{5}{2}$	-1
4	-1	$1 - \frac{3}{2} = -\frac{1}{2}$	$2 - 1 - \frac{7}{2} = -\frac{5}{2}$	-1

Both classifiers have 0 training errors.

Accuracy alone cannot tell us which boundary has more clearance from the observations.

Solution 1(b): Geometric margins

(b) Divide each signed score by the weight norm.
For A, $\boldsymbol{w}_A = (1, 0)^\mathsf{T}$, so $\|\boldsymbol{w}_A\|_2 = 1$.
Its signed scores are $(3/2, 1/2, 3/2, 1/2)$, hence:

$$\boxed{\gamma_{\min,A} = \frac{1/2}{1} = \frac{1}{2}.}$$

For B, $\boldsymbol{w}_B = (2, 1)^\mathsf{T}$ and $\|\boldsymbol{w}_B\|_2 = \sqrt{5}$.
Its signed scores are $(7/2, 5/2, 5/2, 5/2)$, hence:

$$\boxed{\gamma_{\min,B} = \frac{5}{2\sqrt{5}} = \frac{\sqrt{5}}{2} \approx 1.118.}$$

Therefore $\gamma_{\min,B} > \gamma_{\min,A}$: **B has the larger geometric margin**, despite identical training accuracy.

Solution 1(c): Canonical separator

(c) Make the minimum functional margin equal to 1.
For B, $\min_i y_i f_B(\boldsymbol{x}_i) = 5/2$. Divide all coefficients by $5/2$:

$$\boxed{f_{B,c}(\boldsymbol{x}) = \frac{4}{5}x_1 + \frac{2}{5}x_2 - \frac{7}{5}.}$$

The canonical margin lines satisfy $f_{B,c} = \pm 1$:

$$f_{B,c} = +1 \iff \boxed{2x_1 + x_2 = 6},$$
$$f_{B,c} = -1 \iff \boxed{2x_1 + x_2 = 1}.$$

Since $\|\boldsymbol{w}_{B,c}\|_2 = \sqrt{(4/5)^2 + (2/5)^2} = 2/\sqrt{5}$,

$$\boxed{\text{Full margin width} = \frac{2}{\|\boldsymbol{w}_{B,c}\|_2} = \sqrt{5} \approx 2.236.}$$

Solution 1(d): Margin-contact points

(d) Test the canonical margin equations.

i	$2x_{i1} + x_{i2}$	Margin line
1	7	Outside the positive margin
2	6	$f_{B,c} = +1$
3	1	$f_{B,c} = -1$
4	1	$f_{B,c} = -1$

Thus $\boxed{\boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4}$ touch the two margin lines.
For observation 2, $y_2 f_B(\boldsymbol{x}_2) = 5/2$.
Multiplying the entire score by 4 gives:

$$\gamma_2' = \frac{4(5/2)}{\|4(2,1)\|_2} = \frac{10}{4\sqrt{5}} = \boxed{\frac{\sqrt{5}}{2} = \gamma_2}.$$

The **geometric** margin is invariant to positive score rescaling.

Task 2: Hard-margin optimality

A two-dimensional training set has the following classes:

$$\begin{array}{l|l} y=-1 & (0,0),\ (-1,0),\ (0,-1) \\ y=+1 & (2,2),\ (3,2),\ (2,3) \end{array}$$

All coordinates denote column vectors in $\mathbb{R}^2$. Use $f(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{x} + b$ and the constraints $y_i f(\boldsymbol{x}_i) \geq 1$.

(a) The closest opposite-class pair is $(0,0)$ and $(2,2)$. Use their midpoint and connecting direction to find a canonically scaled candidate $\boldsymbol{w}, b$.

(b) Compute signed scores for all six observations and identify the margin-contact points.

(c) Compute the one-sided and full geometric margins and the primal objective $\|\boldsymbol{w}\|_2^2/2$.

(d) Prove the candidate is optimal by subtracting the two closest-point constraints and applying Cauchy–Schwarz.

Solution 2(a): Canonical candidate

(a) Use the direction and midpoint of the closest pair.
The difference from $(0,0)$ to $(2,2)$ is $(2,2)$, so the normal direction is $(1,1)$.
Their midpoint is $(1,1)$; therefore propose:

$$f(\boldsymbol{x}) = t(x_1 + x_2 - 2), \qquad t > 0.$$

At the positive point $(2,2)$, impose the canonical condition $f(2,2) = +1$:

$$t(2+2-2) = 2t = 1 \quad \Longrightarrow \quad t = \frac{1}{2}.$$

Hence:

$$\boxed{\boldsymbol{w} = (1/2, 1/2)^{\mathsf{T}}, \qquad b = -1,} \qquad \boxed{f(\boldsymbol{x}) = \frac{1}{2}x_1 + \frac{1}{2}x_2 - 1.}$$

The negative point $(0,0)$ has score -1, so it also meets its canonical margin.

Solution 2(b): Feasibility and support vectors

(b) Verify every signed score $m_i = y_i f(\boldsymbol{x}_i)$.

$\boldsymbol{x}_i$	y_i	$f(\boldsymbol{x}_i)$	m_i
$(0, 0)$	-1	-1	1
$(-1, 0)$	-1	$-3/2$	$3/2$
$(0, -1)$	-1	$-3/2$	$3/2$
$(2, 2)$	$+1$	1	1
$(3, 2)$	$+1$	$3/2$	$3/2$
$(2, 3)$	$+1$	$3/2$	$3/2$

Every $m_i \geq 1$, so the candidate is feasible.

Margin-contact points: $(0, 0)$ and $(2, 2)$.

The other points lie strictly beyond the margin. Once optimality is established, the two contact points are the support vectors.

Solution 2(c): Margin and objective

(c) Compute the weight norm first.

$$\|\boldsymbol{w}\|_2 = \sqrt{(1/2)^2 + (1/2)^2} = \frac{1}{\sqrt{2}}.$$

The distance from either margin line to the decision boundary is:

$$\boxed{\text{One-sided margin} = \frac{1}{\|\boldsymbol{w}\|_2} = \sqrt{2}.}$$

Therefore:

$$\boxed{\text{Full margin width} = \frac{2}{\|\boldsymbol{w}\|_2} = 2\sqrt{2}.}$$

The hard-margin primal objective is:

$$\boxed{\frac{1}{2}\|\boldsymbol{w}\|_2^2 = \frac{1}{2}\left(\frac{1}{2}\right) = \frac{1}{4}.}$$

Solution 2(d): Proof of optimality

(d) Any feasible separator must satisfy the two closest-point constraints.
For the positive point $(2, 2)$ and negative point $(0, 0)$:

$$\boldsymbol{w}^{\mathsf{T}}(2, 2) + b \geq 1, \qquad \boldsymbol{w}^{\mathsf{T}}(0, 0) + b \leq -1.$$

Subtract to eliminate b:

$$\boldsymbol{w}^{\mathsf{T}}(2, 2) \geq 2.$$

By Cauchy–Schwarz:

$$2 \leq \boldsymbol{w}^{\mathsf{T}}(2, 2) \leq \|\boldsymbol{w}\|_2 \, \|(2, 2)\|_2 = 2\sqrt{2} \, \|\boldsymbol{w}\|_2.$$

Thus $\|\boldsymbol{w}\|_2 \geq 1/\sqrt{2}$, so every feasible solution has $\frac{1}{2}\|\boldsymbol{w}\|_2^2 \geq \frac{1}{4}$.
Our candidate achieves $\boxed{1/4}$; hence it is $\boxed{\text{globally optimal}}$.

Task 3: Soft-margin trade-off

For one-dimensional observations, compare two candidate classifiers:

x_i	-2	-0.5	0.2	2
y_i	-1	-1	$+1$	$+1$

$$f_A(x) = x + 0.1, \qquad f_B(x) = 0.5x + 0.5.$$

Use signed scores $m_i = y_i f(x_i)$, minimum slack $\xi_i = \max(0, 1 - m_i)$, and $J = \frac{1}{2}w^2 + C\sum_i \xi_i$, where $C > 0$.

(a) Compute the four signed scores and minimum slacks for both candidates.

(b) Express $J_A(C)$ and $J_B(C)$ exactly.

(c) Compare the candidates at $C = 0.2$ and $C = 1$.

(d) Find the value of C at which their objectives are equal. Explain the trade-off, including the error made by one candidate.

Solution 3(a): Classifier A

(a) First use $f_A(x) = x + 0.1$.
Compute $f_A(x_i)$, signed scores $m_i = y_i f_A(x_i)$, and slack $\xi_i = \max(0, 1 - m_i)$:

x_i	y_i	$f_A(x_i)$	m_i	ξ_i
-2	-1	-1.9	1.9	0
-0.5	-1	-0.4	0.4	0.6
0.2	$+1$	0.3	0.3	0.7
2	$+1$	2.1	2.1	0

Thus:

$$\boxed{\sum_i \xi_i^{(A)} = 0 + 0.6 + 0.7 + 0 = 1.3.}$$

All four predictions are correct, but the two central observations lie *inside* the desired margin and incur hinge loss.

Solution 3(a): Classifier B

(a) Now use $f_B(x) = 0.5x + 0.5$.

x_i	y_i	$f_B(x_i)$	m_i	ξ_i
-2	-1	-0.5	0.5	0.5
-0.5	-1	0.25	-0.25	1.25
0.2	$+1$	0.6	0.6	0.4
2	$+1$	1.5	1.5	0

Thus:

$$\boxed{\sum_i \xi_i^{(B)} = 0.5 + 1.25 + 0.4 + 0 = 2.15.}$$

Unlike A, classifier B **misclassifies** the negative observation at $x = -0.5$.
Its smaller weight nevertheless reduces the norm penalty.

Solution 3(b): Objective functions

(b) Include the weight penalty and the total slack.

For A, $w_A = 1$ and $\sum_i \xi_i^{(A)} = 1.3$:

$$\boxed{J_A(C) = \tfrac{1}{2}(1)^2 + 1.3C = 0.5 + 1.3C.}$$

For B, $w_B = 0.5$ and $\sum_i \xi_i^{(B)} = 2.15$:

$$\boxed{J_B(C) = \tfrac{1}{2}(0.5)^2 + 2.15C = 0.125 + 2.15C.}$$

- A has the larger norm penalty but the smaller violation penalty.
- B has the smaller norm penalty but more margin violations.

The preferred candidate therefore depends on C.

Solution 3(c): Comparing two costs

(c) Evaluate the same two candidates at different values of C.

C	$J_A(C)$	$J_B(C)$	Lower objective
0.2	$0.5 + 0.2(1.3) = 0.760$	$0.125 + 0.2(2.15) = 0.555$	$\boxed{B}$
1	$0.5 + 1.3 = 1.800$	$0.125 + 2.15 = 2.275$	$\boxed{A}$

For small C, B's smaller weight norm outweighs its extra hinge loss.
For larger C, A's smaller hinge loss becomes more important.
These are comparisons of two specified candidates, not a claim that either is the globally optimal SVM for the dataset.

Solution 3(d): Switching value of C

(d) Set the two objectives equal.

$$0.5 + 1.3C = 0.125 + 2.15C \quad \Longrightarrow \quad 0.375 = 0.85C.$$

Hence:

$$\boxed{C^* = \frac{0.375}{0.85} = \frac{15}{34} \approx 0.441.}$$

- $0 < C < C^*$: candidate B has the smaller objective.
- $C > C^*$: candidate A has the smaller objective.
- $C = C^*$: the two candidates tie.

B makes one training error at $x = -0.5$; A makes none.
A smaller C can still favor B because its weight is smaller.

Task 4: Dual and support vectors

Consider three two-dimensional training observations:

i	$\boldsymbol{x}_i$	y_i
1	$(-1, 0)^\mathsf{T}$	-1
2	$(0, -1)^\mathsf{T}$	-1
3	$(1, 1)^\mathsf{T}$	$+1$

The hard-margin dual maximizes $W(\boldsymbol{\alpha}) = \sum_i \alpha_i - \frac{1}{2}\| \sum_i \alpha_i y_i \boldsymbol{x}_i \|_2^2$, with $\alpha_i \geq 0$ and $\sum_i \alpha_i y_i = 0$.

(a) Set $a = \alpha_1$, $c = \alpha_2$. Use the equality constraint to express α_3 and the primal weight $\boldsymbol{w}$ in terms of a, c.

(b) Reduce W to two variables and find all optimal multipliers.

(c) Recover $\boldsymbol{w}, b$, verify all training margins, and identify the support vectors.

(d) Compute the full margin width and classify $(1, 0)^\mathsf{T}$ and $(0, 0)^\mathsf{T}$.

Solution 4(a): Equality constraint

(a) Use $\sum_i \alpha_i y_i = 0$.
The labels are $(-1, -1, +1)$, so:

$$-\alpha_1 - \alpha_2 + \alpha_3 = 0 \quad \Longrightarrow \quad \boxed{\alpha_3 = a + c.}$$

The primal weight is a weighted sum of the labeled points:

$$\begin{aligned} \boldsymbol{w} &= a(-1)(-1, 0)^{\mathsf{T}} + c(-1)(0, -1)^{\mathsf{T}} + (a + c)(+1)(1, 1)^{\mathsf{T}} \\ &= (a, 0)^{\mathsf{T}} + (0, c)^{\mathsf{T}} + (a + c, a + c)^{\mathsf{T}}. \end{aligned}$$

Therefore:

$$\boxed{\boldsymbol{w} = (2a + c,\ a + 2c)^{\mathsf{T}}}, \qquad a, c \geq 0.$$

This eliminates one dual variable and expresses the decision normal entirely through a, c.

Solution 4(b): Reduced objective

(b) Substitute the expression for $\boldsymbol{w}$.
Because $\alpha_3 = a + c$, the sum of multipliers is $2(a + c)$.
Also:

$$\begin{aligned}\|\boldsymbol{w}\|_2^2 &= (2a + c)^2 + (a + 2c)^2 \\ &= 5a^2 + 8ac + 5c^2.\end{aligned}$$

Thus the dual objective becomes:

$$\boxed{W(a, c) = 2(a + c) - \frac{1}{2}(5a^2 + 8ac + 5c^2).}$$

To maximize it, differentiate with respect to a and c.

Solution 4(b): Optimal multipliers

(b) Solve the two stationary equations.

$$\frac{\partial W}{\partial a} = 2 - 5a - 4c = 0,$$
$$\frac{\partial W}{\partial c} = 2 - 4a - 5c = 0.$$

Subtracting yields $a = c$; substituting gives $9a = 2$.
Therefore:

$$\boxed{a = c = \frac{2}{9}, \qquad \alpha_3 = a + c = \frac{4}{9}.}$$

All are nonnegative. The Hessian of W is $-\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$, whose eigenvalues are $-9, -1$; therefore W is strictly concave and this feasible stationary point is its unique maximum.

Solution 4(c): Recover the classifier

(c) First calculate w using $a = c = 2/9$.

$$\boxed{\boldsymbol{w} = \left(2\frac{2}{9} + \frac{2}{9}, \frac{2}{9} + 2\frac{2}{9}\right)^{\mathsf{T}} = \left(\frac{2}{3}, \frac{2}{3}\right)^{\mathsf{T}}.}$$

Since $\alpha_3 > 0$, the positive observation $(1, 1)$ lies on the canonical margin:

$$\boldsymbol{w}^{\mathsf{T}}(1,1) + b = 1 \quad \Longrightarrow \quad \frac{4}{3} + b = 1.$$

Hence:

$$\boxed{b = -\frac{1}{3}}, \qquad \boxed{f(\boldsymbol{x}) = \frac{2}{3}x_1 + \frac{2}{3}x_2 - \frac{1}{3}.}$$

The resulting boundary is $\boxed{2x_1 + 2x_2 - 1 = 0}$.

Solution 4(c): Verify support vectors

(c) Check the canonical signed score of every observation.

i	$\boldsymbol{x}_i$	$f(\boldsymbol{x}_i)$	$y_i f(\boldsymbol{x}_i)$
1	$(-1, 0)$	$-2/3 - 1/3 = -1$	1
2	$(0, -1)$	$-2/3 - 1/3 = -1$	1
3	$(1, 1)$	$2/3 + 2/3 - 1/3 = 1$	1

All constraints hold with equality. Furthermore, $\alpha_1 = \alpha_2 = 2/9 > 0$ and $\alpha_3 = 4/9 > 0$.

All three observations are support vectors.

The dual uses all three because each is needed in the weighted normal-vector representation.

Solution 4(d): Margin and predictions

(d) Compute the geometric margin width.

$$\|\boldsymbol{w}\|_2 = \sqrt{(2/3)^2 + (2/3)^2} = \frac{2\sqrt{2}}{3}, \qquad \boxed{\text{Full width} = \frac{2}{\|\boldsymbol{w}\|_2} = \frac{3}{\sqrt{2}} \approx 2.121.}$$

For $\boldsymbol{x}_A = (1, 0)^\mathsf{T}$:

$$f(\boldsymbol{x}_A) = \frac{2}{3} - \frac{1}{3} = \frac{1}{3} > 0 \quad \Longrightarrow \quad \boxed{\hat{y}_A = +1}.$$

For $\boldsymbol{x}_B = (0, 0)^\mathsf{T}$:

$$f(\boldsymbol{x}_B) = -\frac{1}{3} < 0 \quad \Longrightarrow \quad \boxed{\hat{y}_B = -1}.$$

The primal objective is $\frac{1}{2}\|\boldsymbol{w}\|_2^2 = 4/9$, equal to the optimal dual value: a useful consistency check.

Tutorial 10 complete

Questions and discussion